

Nucleic Acids

Lecturer 8

Presented By

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Chapter V

From gene to Protein

Chapter five: From gene to protein

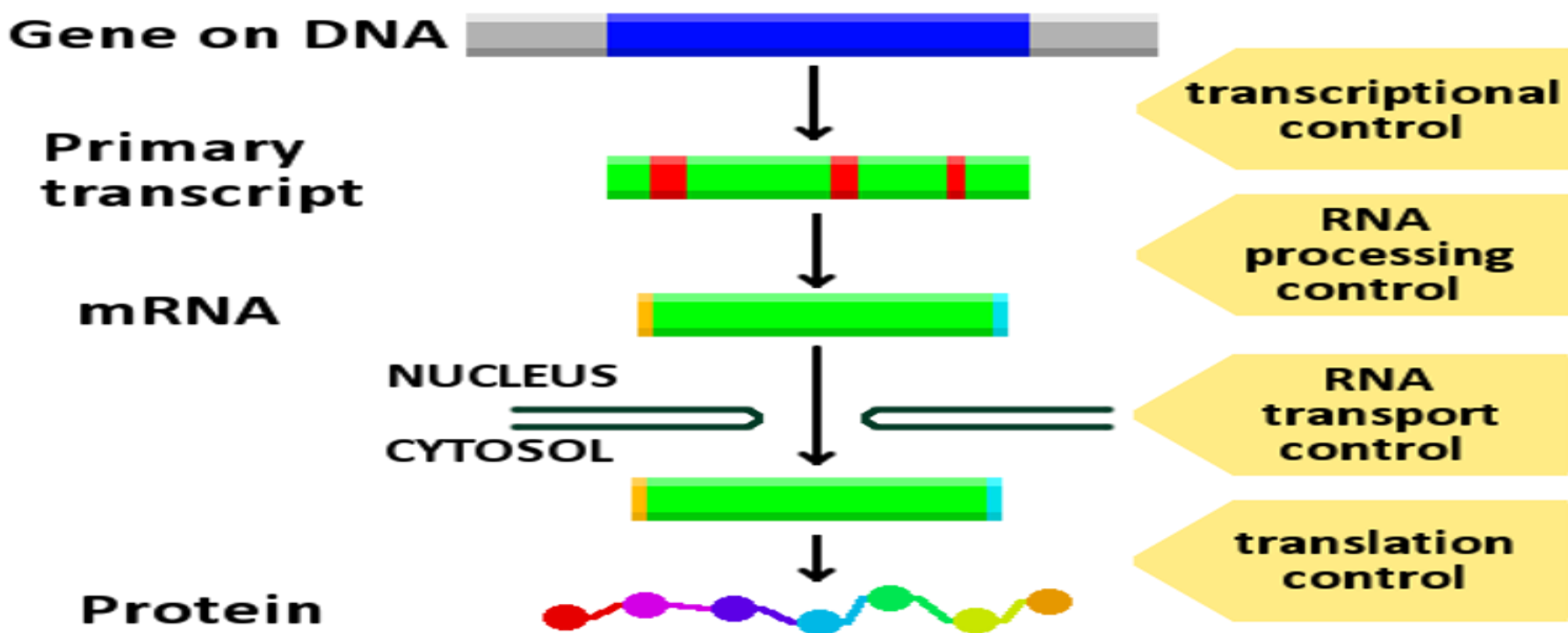
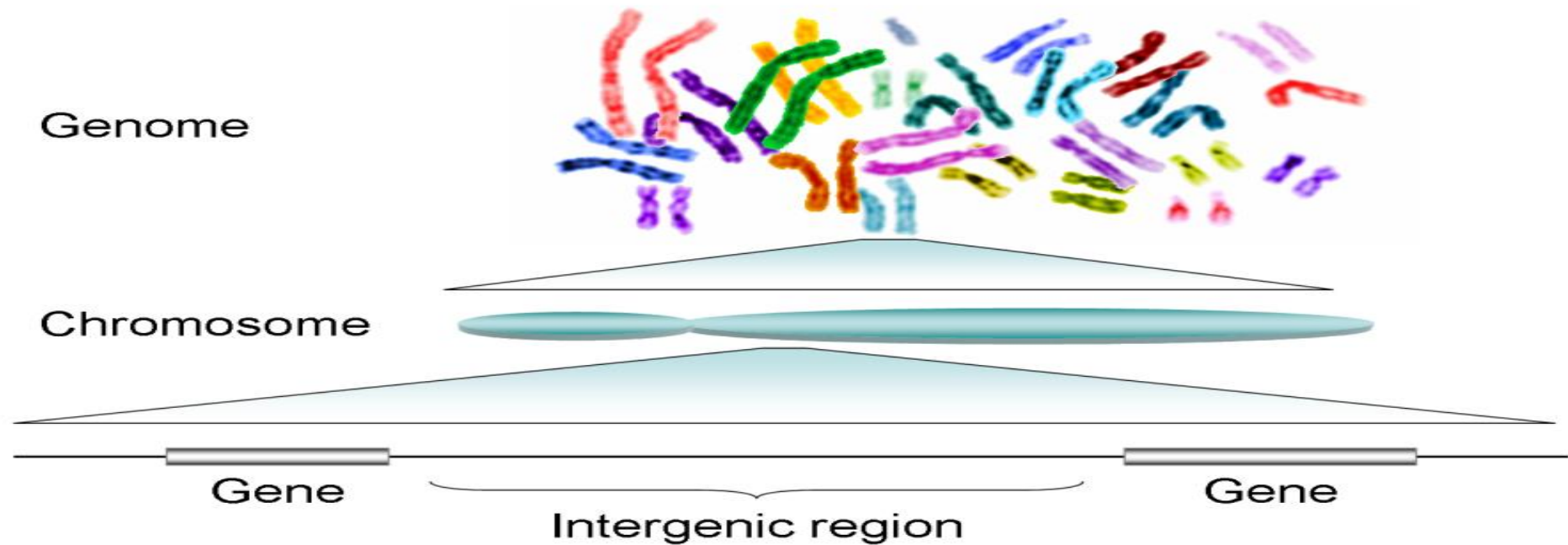
- **1- The cell cycle & DNA Replication**

- **2- Gene Organization & Transcription**

- **3- Genetic code & Protein Translation**

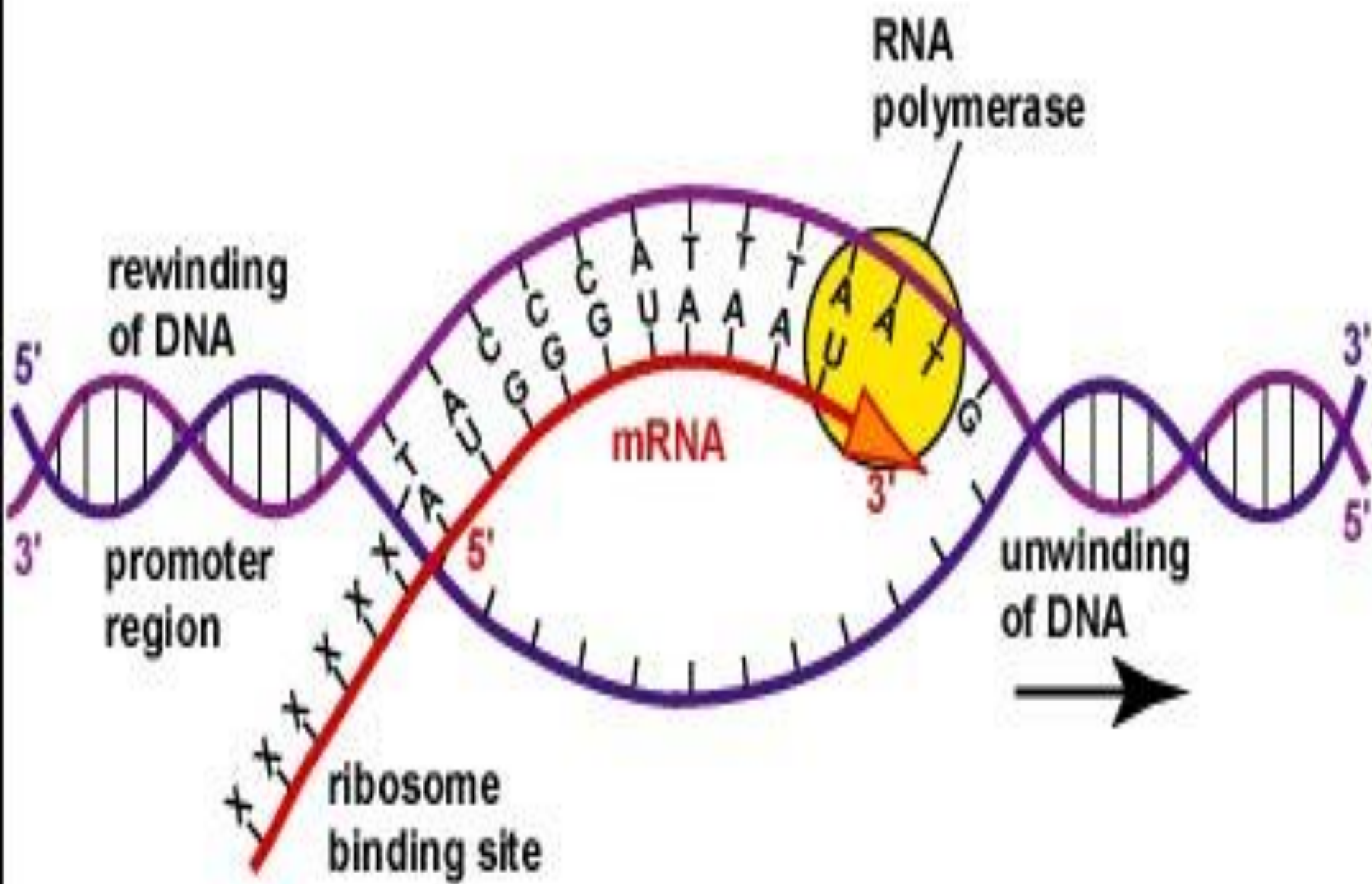
a- DNA and Genes:

- DNA is organized into functional units called “Genes”.
- **Gene** is a segment of DNA (on a specific site on a chromosome) that is responsible for the physical and inheritable characteristics or phenotype of an organism. Genes encode information for making proteins and functional RNA's.
- **Genome:** is the complete DNA sequence of an organism. It consists of DNA (or RNA in RNA viruses). The genome includes both genes (the coding regions) and the non-coding DNA.
- **Gene expression:** is the process by which information from a gene is used in the synthesis of a functional gene product. These products are often proteins, but in non-protein coding genes such as transfer RNA (tRNA) or small nuclear RNA (snRNA) genes, the product is a functional RNA.



b-Transcription:

- is the process of making an RNA copy of a single gene.
- The enzyme used in transcription is "RNA polymerase". There are several forms of RNA polymerase.
- The raw materials for the new RNA are the 4 ribonucleoside triphosphates: ATP, CTP, GTP, and UTP.
- The DNA sequences at which the initiation complex assembles is called a **gene promoter** (are located near the transcription start sites of genes, on the same strand and upstream on the DNA).
- The rate/level of transcription from a given gene is regulated by the activity of **DNA binding proteins**, called **Transcription Factors**. These can be transcriptional activators, or transcriptional repressors, and hence are useful in regulating gene expression.



Chapter Three: From gene to protein

- **1- The cell cycle & DNA Replication**
- **2- Gene Organization & Transcription**
- **3- Genetic code & Protein Translation**

a- Genetic code:

- The nucleotide sequence of the mRNA is composed of four different nucleotides whereas a protein is built up from 20 amino acids.
- To allow the four nucleotides to specify 20 different amino acids, the nucleotide sequence is interpreted in codons, groups of three nucleotides.
- These codons have their corresponding anticodon in the tRNA. Furthermore each anticodon is linked to one particular amino acid. Thus, each codon specifies one amino acid. This is referred to as the genetic code.
- Enzymes called aminoacyl-tRNA synthetases attach the correct amino acid to each tRNA, based on the three-dimensional structure of the tRNA molecule.

Second letter

First letter

	U	C	A	G	
U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G
C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

Third letter

b- Translation:

- is the final step on the way from DNA to protein. It is the synthesis of proteins directed by a mRNA template.
- During translation amino acids are linked together to form a polypeptide chain which will later be folded into a protein.
- Messenger RNA (mRNA) is transcribed from DNA (pre-mRNA) and processed (spliced) in the nucleus.
- mRNA is transported out of the nucleus and translated into protein in the cytoplasm and on the rough endoplasmic reticulum.
- The machinery for translating mRNA is the ribosome, a large 2 subunit complex of proteins and ribosomal RNA (rRNA).
- The translation is dependent on many components, of which two are extra important: tRNA and rRNA.

Amino acids corresponding to the codons are added to the growing protein chain.

